



Princeton Computer Science Contest – Fall 2025

Problem 9: Let It Glow! (20 points)

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The Challenge

Against everyone's advice, you've decided to let randomness determine which Mol Bio lab to do your junior and senior independent work with, and now you're stuck with Princeton's wackiest geneticist. He's given you a strange task: genetically engineer some bacteria to make a protein that glows as bright as possible. You feel a little lost, but thankfully your professor has given you this guidance:

1. You'll be designing a **135**-base-pair genetic sequence (made up of the letters A, T, G, and C).
2. The minimum protein brightness is **0**, and the maximum is **100**.
3. You can test the brightness of up to **100,000** proteins with a microscope before you drive yourself insane.

Submit the DNA sequence that yields brightest protein you can come up with!

Note you can fully solve this problem with NO knowledge of biology. However, some knowledge of biology can be helpful as this problem is inspired by real bacterial operon structures.

Testing brightness

Check out the code template on the website! You'll need to make requests to our server using your team id and your team's **access code**. If you do not have your access code, ping Daniel in the Discord or come to office hours.

Helpful information

In living organisms, proteins are made up of chains of amino acids. In DNA, blocks of 3 base pairs are translated into one amino acid. On the next page is a table mapping each 3-base-pair sequence to the amino acid it encodes.

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	T	C	A	G	
T	TTT } phe	TCT } ser	TAT } tyr	TGT } cys	T
	TTT } phe	TCC } ser	TAC } tyr	TGC } cys	C
	TTA } leu	TCA } ser	TAA } stop	TGA } stop	A
	TTG } leu	TCG } ser	TAG } stop	TGG } trp	G
C	CTT } leu	CCT } pro	CAT } his	CGT } arg	T
	CTC } leu	CCC } pro	CAC } his	CGC } arg	C
	CTA } leu	CCA } pro	CAA } gln	CGA } arg	A
	CTG } leu	CCG } pro	CAG } gln	CGG } arg	G
A	ATT } ile	ACT } thr	AAT } asn	AGT } ser	T
	ATC } ile	ACC } thr	AAC } asn	AGC } ser	C
	ATA } ile	ACA } thr	AAA } lys	AGA } arg	A
	ATG } met	ACG } thr	AAG } lys	AGG } arg	G
G	GTT } val	GCT } ala	GAT } asp	GGT } gly	T
	GTC } val	GCC } ala	GAC } asp	GGC } gly	C
	GTA } val	GCA } ala	GAA } glu	GGA } gly	A
	GTG } val	GCG } ala	GAG } glu	GGG } gly	G

In this table, each three-letter code in bold corresponds to a different amino acid. For this problem, not your entire DNA sequence will get turned into these amino acids — only bases **74-103** (inclusive, zero-indexed). Just like in real life, the rest of the sequence follows different rules that you have to figure out!

Different amino acids have different chemical properties. This problem uses the following classifications:

Hydrophobic	Polar uncharged	Positive	Negative	Special
Ala	Ser	Lys	Asp	Gly
Val	Thr	Arg	Glu	Pro
Ile	Asn	His		
Leu	Gln			
Met	Cys			
Phe				
Trp				
Tyr				

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